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SENSING-AIDED COMMUNICATION CHANNEL ESTIMATION WITH TENSOR-BASED MOVING TARGET LOCALIZATION

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ABSTRACT

In the integrated sensing and communication system, sensing functionalities are expected to benefit the communication instead of compromising its performance. In this paper, a sensing-aided communication channel estimation method is proposed, where the non-cooperative moving targets are localized and the associated propagation paths are excluded from the channel. Specifically, the received signal is formulated as a high-order tensor and then decomposed for channel parameter estimation. The parameters including velocity and angles of each path are automatically paired in the decomposed tensor factors, which enables identification of the high-velocity paths of moving targets. Then, by excluding the parameters of moving targets, a stable communication channel can be constructed. According to simulation, the proposed method contributes to enhanced data transmission performance while accurately localizing the moving targets.

***Index Terms*—** Channel estimation, integrated sensing and communication, moving target localization, tensor.

1. INTRODUCTION

Integrated sensing and communication (ISAC), which facilitates applications including internet of vehicles and internet of drones, will play a crucial role in the next-generation wireless networks [1–4]. Therefore, research efforts have focused on integrating sensing functionalities into existing communication systems [5–7]. In particular, to enable non-cooperative target localization, additional propagation paths associated with these targets are included in the channel matrix to be estimated [8–10]. However, the dynamic nature of these paths impacts the reliability of the estimated channel, leading to degraded communication performance [11, 12]. Thus, it is important to design an ISAC channel estimation which maintains or even enhances communication performance.

The existing parametric channel estimation methods in the millimeter wave (mmWave) frequency band characterize the channel by location-related parameters [13, 14]. To increase communication rate using target sensing in the ISAC system, these estimated parameters are utilized in uplink channel construction, which aims to provide more propagation paths supporting parallel data transmission [15, 16]. Nevertheless, the high-mobility characteristics of moving targets is omitted, which harms the reliability of the estimated channel. In this regard, the paths associated with these targets are supposed to be excluded in communication channel estimation. It is a challenging task within the matrix-based signal processing paradigm due to the coupling of parameters from all paths [17, 18]. The emerging tensor-based channel estimation methods are able to decouple channel parameters in the decomposed factors with automatic pairing [19–21]. However, in these methods, the identification of target-associated propagation paths has not been investigated. Hence, it is still of interest to develop an effective moving target sensing-aided communication channel estimation method tailored for ISAC.

In this paper, we propose a communication channel estimation method aided with tensor-based moving target localization. Specifically, the received signal is modeled as a high-order tensor and then decomposed for channel parameter estimation. Since the parameters of each propagation path are automatically decoupled and paired, the paths with high velocity can be identified to enable moving target localization. Then, the communication channel estimator can be constructed by excluding the paths associated with moving targets. The proposed method enables to enhance data transmission stability and also offers accurate moving target localization results.

2. TENSOR SIGNAL MODEL FOR ISAC

We consider a fully-digital multiple-input multiple-output (MIMO) ISAC system consisting of a base station (BS) with

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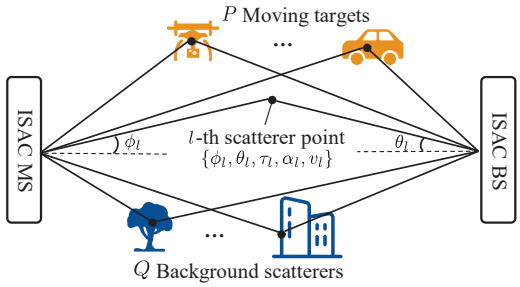

Fig. 1. Illustration of the deployed ISAC system.

N_{BS} antennas and a mobile station (MS) with N_{MS} antennas as shown in Fig. 1. The BS and MS are equipped with a uniform linear array (ULA), whose inter-element spacing d equals to half of the signal wavelength $\lambda = c/f_c$. Here, c denotes the light speed, and f_c denotes the carrier frequency.

The MS and BS communicate using K subcarriers in a frame containing N_{p} subframes, and each subframe is composed of M orthogonal frequency-division multiplexing (OFDM) symbols. In the first subframe, the pilot symbols transmitted from the MS are used for both channel estimation and target localization, while the remaining subframes carry N_{s} ($N_{\mathrm{s}} \leq N_{\mathrm{MS}} \leq N_{\mathrm{BS}}$) independent uplink data streams from the MS. The received signal vector $\boldsymbol{y}_k(m)$ of the k-th subcarrier corresponding to the m-th transmitted OFDM symbol is represented as

$$\begin{aligned} \boldsymbol{y}_k(m) = \boldsymbol{W}^{\mathrm{H}}\boldsymbol{H}_k(m)\boldsymbol{F}\boldsymbol{s}_k(m) + \boldsymbol{W}^{\mathrm{H}}\boldsymbol{n}_k, \\ \forall k = 1, 2, \cdots, K,\ m = 1, 2, \cdots, M, \end{aligned} \tag{1}$$

where $\boldsymbol{F}$ and $\boldsymbol{W}$ respectively denote the precoding matrix of the MS and combining matrix of the BS, $\boldsymbol{s}_k(m)$ is the complex symbol vector, $\boldsymbol{H}_k(m)$ is the channel matrix, $\boldsymbol{n}_k \sim \mathcal{CN}\left(\boldsymbol{0}, \sigma_{\mathrm{n}}^2\boldsymbol{I}\right)$ is the additive Gaussian white noise vector with σ_{n}^2 being the noise power, and $[\cdot]^{\mathrm{H}}$ denotes the Hermitian transpose operator.

Due to the sparsity of mmWave channel, there are finite number of resolvable paths between the MS and BS. These paths are either generated by the moving targets or quasi-stationary environmental objects serving as scatterer points[1], while only the latter contribute to communication channel estimation for stable data transmission. Assume that there are L dominant paths, among which P paths are associated with the moving non-cooperative targets with high bistatic velocity $|v_p| \geq |\xi|, p = 1, 2, \cdots, P$, while the rest $Q = L - P$ paths are associated with the environmental objects with $|v_q| < |\xi|, q = P+1, P+2, \cdots, L$. Here, ξ denotes an empirically determined threshold, and $|\cdot|$ denotes the absolute value of a number. Suppose that the MS remains stationary and the Doppler shift mainly comes from the scatterer points. Then, the time-varying channel matrix can be represented as

$$\boldsymbol{H}_k(m) = \sum_{l=1}^{L} \alpha_l e^{\jmath 2\pi m T_{\mathrm{s}} f_l} e^{-\jmath 2\pi k \tau_l \Delta} \boldsymbol{a}_{\mathrm{BS}}(\theta_l)\boldsymbol{a}_{\mathrm{MS}}^{\mathrm{H}}(\phi_l), \tag{2}$$

where α_l is the complex path gain, $\tau_l = (R_{\mathrm{MS},l} + R_{\mathrm{BS},l})/c$ is the propagation delay, $R_{\mathrm{MS},l}$ is the distance between the l-th scatterer point and the MS, and $R_{\mathrm{BS},l}$ is the distance between the l-th scatterer point and the BS, Δ denotes the subcarrier spacing, T_{s} denotes the elementary symbol duration including the cyclic prefix,

$$\begin{aligned} \boldsymbol{a}_{\mathrm{BS}}(\theta_l) &= [1, e^{-\jmath\pi \sin\theta_l}, \cdots, e^{-\jmath\pi(N_{\mathrm{BS}}-1)\sin\theta_l}]^{\mathrm{T}}, \\ \boldsymbol{a}_{\mathrm{MS}}(\phi_l) &= [1, e^{-\jmath\pi \sin\phi_l}, \cdots, e^{-\jmath\pi(N_{\mathrm{MS}}-1)\sin\phi_l}]^{\mathrm{T}} \end{aligned} \tag{3}$$

respectively denote the steering vectors of the arrays at the BS and MS with θ_l and ϕ_l being the angle-of-arrival (AOA) and angle-of-departure (AOD), $[\cdot]^{\mathrm{T}}$ denotes the transpose operator,

$$f_l = \frac{1}{\lambda}\frac{\mathrm{d}\left[R_{\mathrm{MS},l} + R_{\mathrm{BS},l}\right]}{\mathrm{d}t} = \frac{2v_l}{\lambda}\sin\left(\frac{|\phi_l| + |\theta_l|}{2}\right) \tag{4}$$

denotes the Doppler frequency of the l-th path, $l = 1, 2, \cdots, L$, and $\jmath = \sqrt{-1}$. Specifically, θ_l, ϕ_l, f_l, and τ_l change continuously between individual symbol periods due to the velocities of scatterer points, especially those of moving targets, inducing considerable channel fluctuation within the entire frame.

During the pilot transmission stage, however, the variance of θ_l, ϕ_l, f_l, and τ_l are neglected due to the relatively brief duration [22]. The precoding matrix $\boldsymbol{F}$ and combining matrix $\boldsymbol{W}$ are both set to identity matrices, and the pilots remain the same for different subcarriers, i.e., $\boldsymbol{s}_k(m) = \boldsymbol{s}(m)$, $\forall k = 1, 2, \cdots, K$. To preserve the multi-dimensional structure of the uplink signal, the signal can be modeled in a tensor form [23]. The tensor decomposition facilitates decoupled and paired channel parameter estimation. Specifically, by substituting (2) into (1) and then concatenating the receive signal vectors from all the K subcarriers and M symbols, a third-order tensor can be formulated as

$$\boldsymbol{\mathcal{Y}} = \sum_{l=1}^{L} \tilde{\boldsymbol{a}}_{\mathrm{BS}}(\theta_l) \circ [\alpha_l \boldsymbol{u}(\tau_l)] \circ \left[\boldsymbol{v}(f_l) \odot (\boldsymbol{S}^{\mathrm{T}}\tilde{\boldsymbol{a}}_{\mathrm{MS}}(\phi_l))\right] + \boldsymbol{\mathcal{N}}, \tag{5}$$

where $\tilde{\boldsymbol{a}}_{\mathrm{BS}}(\theta_l) = \boldsymbol{W}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{BS}}(\theta_l)$, $\tilde{\boldsymbol{a}}_{\mathrm{MS}}(\phi_l) = \boldsymbol{F}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{MS}}(\phi_l)$, $\boldsymbol{S} = [\boldsymbol{s}(1), \boldsymbol{s}(2), \cdots, \boldsymbol{s}(M)]$ is the pilot symbol matrix, $\boldsymbol{u}(\tau_l) = [e^{-\jmath 2\pi\tau_l\Delta}, e^{-\jmath 2\pi 2\tau_l\Delta}, \cdots, e^{-\jmath 2\pi K\tau_l\Delta}]^{\mathrm{T}}$ is the propagation delay vector, $\boldsymbol{v}(f_l) = [e^{\jmath 2\pi T_{\mathrm{s}} f_l}, e^{\jmath 2\pi 2T_{\mathrm{s}} f_l}, \cdots, e^{\jmath 2\pi M T_{\mathrm{s}} f_l}]^{\mathrm{T}}$ is the Doppler shift vector, $\boldsymbol{\mathcal{N}}$ is the noise tensor, and $\circ$ and $\odot$ denote the outer product and Hadamard product, respectively.

3. CHANNEL ESTIMATION AIDED WITH MOVING TARGET LOCALIZATION

3.1. Tensor-based Moving Target Localization

Since the formulated tensor signal $\boldsymbol{\mathcal{Y}}$ in (5) obeys the typical rank-L canonical polyadic (CP) model, it can be decom-

[1]The line-of-sight (LOS) path does not contain a scatterer point. Nevertheless, we assume that an equivalent scatterer point is located somewhere on the LOS path between BS and MS to treat it as the other paths.

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posed to obtain the closed-form expression for the constituting CP factors, i.e., $\boldsymbol{b}_1(l)=\tilde{\boldsymbol{a}}_{\rm BS}(\hat{\theta}_l)$, $\boldsymbol{b}_2(l)=\hat{\alpha}_l\boldsymbol{u}(\hat{\tau}_l)$, and $\boldsymbol{b}_3(l)=\boldsymbol{v}(\hat{f}_l)\odot\left(\boldsymbol{S}^{\rm T}\tilde{\boldsymbol{a}}_{\rm MS}(\hat{\phi}_l)\right)$, where $\hat{\alpha}_l,\hat{\theta}_l,\hat{\tau}_l,\hat{f}_l,\hat{\phi}_l$ are the estimations of $\alpha_l,\theta_l,\tau_l,f_l,\phi_l$, respectively. The CP decomposition of $\boldsymbol{\mathcal{Y}}$ can then be formulated as the least squares optimization problem

$$\min_{\boldsymbol{B}_1,\boldsymbol{B}_2,\boldsymbol{B}_3}\|\boldsymbol{\mathcal{Y}}-[\![\boldsymbol{B}_1,\boldsymbol{B}_2,\boldsymbol{B}_3]\!]\|_{\rm F}^2\,, \tag{6}$$

where $\boldsymbol{B}_1=[\boldsymbol{b}_1(1),\boldsymbol{b}_1(2),\cdots,\boldsymbol{b}_1(L)]$, $\boldsymbol{B}_2=[\boldsymbol{b}_2(1),\boldsymbol{b}_2(2),\cdots,\boldsymbol{b}_2(L)]$, and $\boldsymbol{B}_3=[\boldsymbol{b}_3(1),\boldsymbol{b}_3(2),\cdots,\boldsymbol{b}_3(L)]$ are the estimated factor matrices, and $\|\cdot\|_{\rm F}$ denotes the Frobenius norm. The above CP decomposition problem can be solved by the alternating least squares (ALS) algorithm[2], which minimizes the fitting error by iteratively updating the estimated factor matrices.

To achieve reliable communication, we aim to exclude the paths associated with moving targets, and estimate the communication channel that benefits the communication performance only. Since the parameters corresponding to each scatterer point are automatically paired in the estimated CP factors, the paths of moving targets can be distinguished through velocity estimation. Meanwhile, the moving target can be localized via joint angle and distance estimation. Specifically, the AOA and propagation delay of the l-th scatterer point are respectively estimated as

$$\hat{\theta}_l=\arcsin\frac{\sum_{n_{\rm BS}=1}^{N_{\rm BS}-1}\angle\left([\boldsymbol{B}_1]_{(n_{\rm BS},l)}/[\boldsymbol{B}_1]_{(n_{\rm BS}+1,l)}\right)}{\pi(N_{\rm BS}-1)} \tag{7}$$

and

$$\hat{\tau}_l=\frac{\sum_{k=1}^{K-1}\angle\left([\boldsymbol{B}_2]_{(k,l)}/[\boldsymbol{B}_2]_{(k+1,l)}\right)}{2(K-1)\pi\Delta}, \tag{8}$$

where $[\boldsymbol{B}_1]_{(n_{\rm BS},l)}$ represents the $(n_{\rm BS},l)$-th element of $\boldsymbol{B}_1$. The total length of the l-th channel path can then be calculated as $\hat{d}_l=c\hat{\tau}_l$.

Since the distance between the MS and BS $d_{\rm bis}$ can be determined by the time-of-flight measurement or Global Positioning System, it can be used as the priori to uniquely calculate the AOD $\hat{\phi}_l$ based on the estimated AOA $\hat{\theta}_l$ and the length of the path $\hat{d}_l$. According to the geometric spatial relationship, the AOD $\hat{\phi}_l$ can be calculated as

$$\hat{\phi}_l=\arccos\left(\frac{\hat{d}_l^2+g_l^2-d_{\rm bis}^2}{2\hat{d}_l g_l}\right)-\arccos\left(\frac{d_{\rm bis}^2+g_l^2-\hat{d}_l^2}{2d_{\rm bis}g_l}\right), \tag{9}$$

where $g_l=\sqrt{\hat{d}_l^2+d_{\rm bis}^2-2\hat{d}_l d_{\rm bis}\cos\hat{\theta}_l}$ represents the distance between the MS and a certain auxiliary point located at a distance $\hat{d}_l$ in direction of the AOA $\hat{\theta}_l$ relative to the BS. Then, the Doppler shift of the l-th scatterer point can be estimated from $\boldsymbol{B}_3$ as

$$\hat{f}_l=\frac{\sum_{m=1}^{M-1}\angle\left(\frac{[\boldsymbol{B}_3]_{(m+1,l)}}{([\boldsymbol{S}]_{(\cdot,m+1)}^{\rm T}\tilde{\boldsymbol{a}}_{\rm MS}(\hat{\phi}_l))}\Big/\frac{[\boldsymbol{B}_3]_{(m,l)}}{([\boldsymbol{S}]_{(\cdot,m)}^{\rm T}\tilde{\boldsymbol{a}}_{\rm MS}(\hat{\phi}_l))}\right)}{2(M-1)\pi T_{\rm s}\Delta}, \tag{10}$$

where $[\boldsymbol{S}]_{(\cdot,m)}$ stands for the m-th column of $\boldsymbol{S}$.

Substituting (4) into (10) yields the bistatic velocity of the l-th scatterer point as

$$\hat{v}_l=\frac{\lambda\hat{f}_l}{2\sin((|\hat{\theta}_l|+|\hat{\phi}_l|)/2)}. \tag{11}$$

By comparing the estimated velocity $\hat{v}_l$ with the threshold ξ as defined in Section 2, we can now determine whether the l-th scatterer point is a moving target or a background scatterer. The position of the p-th moving target relative to the BS in a Cartesian coordinate system can be calculated as

$$\begin{bmatrix}\hat{x}_p\\ \hat{y}_p\end{bmatrix}=\hat{d}_p\frac{\sin(|\hat{\phi}_p|)}{\sin(|\hat{\theta}_p|+|\hat{\phi}_p|)}\begin{bmatrix}\sin(\hat{\theta}_p)\\ \cos(\hat{\theta}_p)\end{bmatrix}. \tag{12}$$

Subsequently, the of these targets will be leveraged to construct a reliable communication channel.

3.2. Communication Channel Estimation

Since the trajectory of moving targets exhibits high variability, the parameters of the associated paths change rapidly within the entire frame, which becomes harmful to the data transmission. In this regard, we aim to enhance communication performance by mitigating these path effects, and only utilize the paths associated with slowly varying background to yield the communication channel

$$\hat{\boldsymbol{H}}_k(\acute{m})=\sum_q\hat{\alpha}_q e^{\jmath 2\pi\acute{m}T_{\rm s}\hat{f}_q}e^{-\jmath 2\pi k\hat{\tau}_q\Delta}\boldsymbol{a}_{\rm BS}(\hat{\theta}_q)\boldsymbol{a}_{\rm MS}^{\rm H}(\hat{\phi}_q), \tag{13}$$

where $\acute{m}\in[M+1,MN_{\rm p}]$ denotes the index of OFDM symbol during data transmission, and the complex channel gain can be estimated as $\hat{\alpha}_q=\gamma^{(1)}\gamma^{(2)}\gamma^{(3)}$. Here, $\gamma^{(1)}$, $\gamma^{(2)}$, and $\gamma^{(3)}$ are the respective mean values of diagonal elements in

$$\begin{aligned}\boldsymbol{\Gamma}_1&=[\boldsymbol{B}_1]_{(\cdot,q)}\boldsymbol{a}_{\rm BS}^{\rm H}(\hat{\theta}_q),\\ \boldsymbol{\Gamma}_2&=[\boldsymbol{B}_2]_{(\cdot,q)}\boldsymbol{u}^{\rm H}(\hat{\tau}_q)\,,\\ \boldsymbol{\Gamma}_3&=[\boldsymbol{B}_3]_{(\cdot,q)}\left(\boldsymbol{v}(\hat{f}_q)\odot(\boldsymbol{S}^{\rm T}\boldsymbol{a}_{\rm MS}(\hat{\phi}_q))\right)^{\rm H}.\end{aligned} \tag{14}$$

As such, the velocities of the localized moving targets are exploited and only the beneficial paths for communication channel estimation are preserved, thereby ensuring stable data transmission.

Based on the estimated communication channel $\hat{\boldsymbol{H}}_k(\acute{m})$ in (13), the precoding and combining matrices corresponding to the $\acute{m}$-th data symbol, for example, can be respectively

[2]The ALS is employed due to its conceptual simplicity, while other algorithms such as the structure-constrained tensor decomposition can also be utilized [24, 25].

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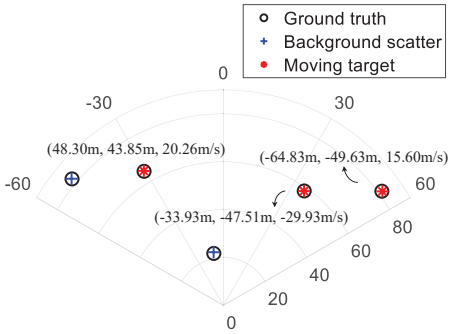

Fig. 2. Performance of moving target localization.

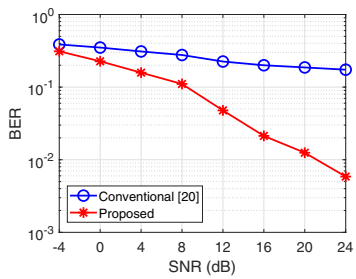

Fig. 3. Comparison of communication performance.

calculated as $\boldsymbol{W}(\acute{m}) = [\boldsymbol{U}]_{(\cdot,[1\ldots\min(N_\mathrm{s},Q)])}$ and $\boldsymbol{F}(\acute{m}) = [\boldsymbol{V}]_{(\cdot,[1\ldots\min(N_\mathrm{s},Q)])}$, where $\boldsymbol{U}$ and $\boldsymbol{V}$ are the unitary matrices from singular value decomposition (SVD)[3] of $\hat{\boldsymbol{H}}_k(\acute{m}) = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\mathrm{H}$. As such, the dominant transmitted energy can be allocated along the paths with slowly-varying channel parameters in data transmission stage, and consequently, the communication reliability can be improved.

4. SIMULATION

We consider a MIMO-OFDM ISAC system whose settings are specified in Table 1. For each scene, 200 Monte-Carlo trials are run, and in each trial $N_\mathrm{s}M(N_\mathrm{p}-1)$ OFDM symbols are randomly generated, which are modulated via quadrature phase-shift keying (QPSK).

We first demonstrate the performance of moving target localization in Fig. 2, where the signal-to-noise ratio (SNR) is set to 0 dB. Consider 5 propagation paths between the BS and MS, including 3 of fast-moving targets and 2 of background scatterers with initial values $[\theta_1, \theta_2, \cdots, \theta_5] = [-30.6°, 35.2°, 54.2°, -50.1°, -10.5°]$, $[\phi_1, \phi_2, \cdots, \phi_5] = [20.9°, -45.1°, -30.1°, 18.6°, 45.2°]$, $[v_1, v_2, \cdots, v_5] = [20.2, -30.0, 15.6, -1.5, 2.3]$m/s, and $[\alpha_1, \alpha_2, \cdots, \alpha_5] = [0.6 + 1.0\jmath, 0.2 + 0.2\jmath, 0.2 - 0.3\jmath, 0.7 - 0.8\jmath, 0.2 + 0.5\jmath]$. Then, θ_l, ϕ_l, f_l, and τ_l are calculated and updated for each symbol duration according to the velocity v_l and the distance between BS and MS d_bis. The results of localization and velocity estimation of moving targets are presented in the form of $(\hat{x}_p, \hat{y}_p, \hat{v}_p)$. It is observed that the moving targets can be distinguished and accurately localized in the pilot transmission stage.

[3]Note that the SVD-based precoding and combining matrix design is utilized as an illustrative example, while other strategies such as zero-forcing and maximum-ratio combining can be employed as well [26, 27].

Table 1. SYSTEM SETTINGS

Parameter	Symbol	Value
Number of transmit elements	N_MS	4
Number of receive elements	N_BS	32
Number of data streams	N_s	2
Carrier frequency	f_c	28 GHz
Subcarrier spacing	Δ	60 kHz
Elementary symbol duration	T_s	17.84 μs
Number of subcarriers	K	100
OFDM symbols per subframe	M	256
Subframes per frame	N_p	10
Distance between BS and MS	d_bis	100 m
Velocity threshold	ξ	5 m/s

Then, to validate the communication stability of the proposed communication channel estimation method, the bit error rate (BER) obtained by using the proposed method is compared to that of the conventional tensor-based channel estimation method [20]. In [20], the channel paths are utilized without distinguishing those associated with moving targets. As shown in Fig. 3, the proposed method exhibits much lower BER compared to the conventional method. The reason is that the channel in data transmission stage can deviate significantly from the estimated channel due to high variability in parameters associated with moving targets, posing threat to reliable data transmission. In contrast, such deviation is prevented by excluding the highly dynamic parameters in the proposed sensing-aided channel estimation method.

Based on the above findings, it is obvious that the sensing capability not only serves as an additional feature in the ISAC system, but also can be adopted to benefit the communication with proper acquisition of the parametric channel information, contributing to more effective data transmission.

5. CONCLUSION

We proposed a sensing-aided communication channel estimation method for ISAC. The channel parameters estimated from tensor decomposition are classified into two groups based on velocity estimation, and only the parameters with low-velocity are utilized to construct stable communication channel. Simulation results validate the effectiveness of the proposed method for enhancing data transmission reliability aided with accurate moving target localization.

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6. REFERENCES

[1] F. Liu, L. Zheng, Y. Cui, C. Masouros, A. P. Petropulu, H. Griffiths, and Y. C. Eldar, "Seventy years of radar and communications: The road from separation to integration," *IEEE Signal Process. Mag.*, vol. 40, no. 5, pp. 106–121, July 2023.

[2] A. Hassanien, M. G. Amin, Y. D. Zhang, and F. Ahmad, "Dual-function radar-communications: Information embedding using sidelobe control and waveform diversity," *IEEE Trans. Signal Process.*, vol. 64, no. 8, pp. 2168–2181, Apr. 2016.

[3] C. Shi, F. Wang, M. Sellathurai, J. Zhou, and S. Salous, "Power minimization-based robust OFDM radar waveform design for radar and communication systems in coexistence," *IEEE Trans. Signal Process.*, vol. 66, no. 5, pp. 1316–1330, Mar. 2018.

[4] Z. Shi, X. Chang, C. Yang, Z. Wu, and J. Wu, "An acoustic-based surveillance system for amateur drones detection and localization," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2731–2739, Mar. 2020.

[5] L. Zheng, M. Lops, Y. C. Eldar, and X. Wang, "Radar and communication coexistence: An overview: A review of recent methods," *IEEE Signal Process. Mag.*, vol. 36, no. 5, pp. 85–99, Sep. 2019.

[6] P. Kumari, S. A. Vorobyov, and R. W. Heath, "Adaptive virtual waveform design for millimeter-wave joint communication-radar," *IEEE Trans. Signal Process.*, vol. 68, pp. 715–730, 2020.

[7] L. Lin, H. Zheng, C. Zhou, S. He, and Z. Shi, "Nonorthogonal waveform assisted DOA estimation for joint MIMO sensing and communication," *EURASIP J. Adv. Signal Process.*, vol. 2023, no. 1, pp. 1–19, Feb. 2023.

[8] J. A. Zhang, M. L. Rahman, K. Wu, X. Huang, Y. J. Guo, S. Chen, and J. Yuan, "Enabling joint communication and radar sensing in mobile networks—A survey," *IEEE Commun. Surv. Tut.*, vol. 24, no. 1, pp. 306–345, Oct. 2022.

[9] X. Chen, Z. Feng, J. A. Zhang, Z. Wei, X. Yuan, and P. Zhang, "Sensing-aided uplink channel estimation for joint communication and sensing," *IEEE Wireless Commun. Lett.*, vol. 12, no. 3, pp. 441–445, Mar. 2023.

[10] L. Pucci, E. Matricardi, E. Paolini, W. Xu, and A. Giorgetti, "Performance analysis of a bistatic joint sensing and communication system," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC)*, Seoul, Korea, May 2022, pp. 73–78.

[11] J. Li, Y. Niu, H. Wu, B. Ai, S. Chen, Z. Feng, Z. Zhong, and N. Wang, "Mobility support for Millimeter wave communications: Opportunities and challenges," *IEEE Commun. Surv. Tut.*, vol. 24, no. 3, pp. 1816–1842, thirdquarter 2022.

[12] R. Zhang, L. Cheng, S. Wang, Y. Lou, W. Wu, and D. W. K. Ng, "Tensor decomposition-based channel estimation for hybrid mmWave massive MIMO in high-mobility scenarios," *IEEE Trans. Commun.*, vol. 70, no. 9, pp. 6325–6340, Sep. 2022.

[13] J. Li, M. F. Da Costa, and U. Mitra, "Joint localization and orientation estimation in Millimeter-wave MIMO OFDM systems via atomic norm minimization," *IEEE Trans. Signal Process.*, vol. 70, pp. 4252–4264, Aug. 2022.

[14] J. Palacios, N. González-Prelcic, and C. Rusu, "Low complexity joint position and channel estimation at millimeter wave based on multidimensional orthogonal matching pursuit," in *Proc. European Signal Process. Conf. (EUSIPCO)*, Belgrade, Serbia, Aug. 2022, pp. 1002–1006.

[15] Z. Huang, K. Wang, A. Liu, Y. Cai, R. Du, and T. X. Han, "Joint pilot optimization, target detection and channel estimation for integrated sensing and communication systems," *IEEE Trans. Wireless Commun.*, vol. 21, no. 12, pp. 10351–10365, Dec. 2022.

[16] R. Mundlamuri, R. Gangula, C. K. Thomas, F. Kaltenberger, and W. Saad, "Sensing aided channel estimation in wideband millimeter-wave MIMO systems," *arXiv:2302.02065*, 2023.

[17] Z. Ni, J. A. Zhang, X. Huang, K. Yang, and J. Yuan, "Uplink sensing in perceptive mobile networks with asynchronous transceivers," *IEEE Trans. Signal Process.*, vol. 69, pp. 1287–1300, 2021.

[18] C. Zhou, Y. Gu, Z. Shi, and M. Haardt, "Structured Nyquist correlation reconstruction for DOA estimation with sparse arrays," *IEEE Trans. Signal Process.*, vol. 71, pp. 1849–1862, Mar. 2023.

[19] Z. Zhou, J. Fang, L. Yang, H. Li, Z. Chen, and R. S. Blum, "Low-rank tensor decomposition-aided channel estimation for Millimeter wave MIMO-OFDM systems," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 7, pp. 1524–1538, July 2017.

[20] J. Wang, W. Zhang, Y. Chen, Z. Liu, J. Sun, and C.-X. Wang, "Time-varying channel estimation scheme for uplink MU-MIMO in 6G systems," *IEEE Trans. Veh. Technol.*, vol. 71, no. 11, pp. 11820–11831, Nov. 2022.

[21] H. Zheng, C. Zhou, Z. Shi, Y. Gu, and Y. D. Zhang, "Coarray tensor direction-of-arrival estimation," *IEEE Trans. Signal Process.*, vol. 71, pp. 1128–1142, Mar. 2023.

[22] Q. Qin, L. Gui, P. Cheng, and B. Gong, "Time-varying channel estimation for millimeter wave multiuser MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 67, no. 10, pp. 9435–9448, Oct. 2018.

[23] H. Zheng, Z. Shi, C. Zhou, and A. L. F. de Almeida, "Coarray tensor completion for DOA estimation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 5472–5486, Oct. 2023.

[24] P. Comon, X. Luciani, and A. L. F. de Almeida, "Tensor decompositions, alternating least squares and other tales," *J. Chemom.*, vol. 23, no. 7/8, pp. 393–405, Apr. 2009.

[25] F. Xu, H. Zheng, and S. A. Vorobyov, "Tensor-based 2D DOA estimation for L-shaped nested array," *IEEE Trans. Aerosp. Electron. Syst.*, pp. 1–16, 2024.

[26] T. L. Marzetta and H. Yang, *Fundamentals of Massive MIMO*, Cambridge University Press, 2016.

[27] H. Zheng, C. Zhou, Z. Shi, and G. Liao, "Sub-Nyquist tensor beamformer: A coprimality constrained design," *IEEE Trans. Signal Process.*, vol. 71, pp. 4163–4177, 2023.

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